

COURSE DESCRIPTION

Course Number	0418-466	Course Title	Introduction to Machine Learning
Credit Hours	3 (3:0:3)	Course Coordinator	Prof. Hazem Raafat
Compulsory	No	Prerequisite	0418-301

Textbook

Introduction to Machine learning, Ethem Alpaydin, MIT Press, Fourth Edition, 2020.
References:
Machine learning, Tom Mitchell, McGraw-Hill, 1997.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction to machine learning theory and tasks	3
2	Statistical-based learning	6
3	Supervised learning	9
4	Deep learning	9
5	Unsupervised learning	6
6	Reinforcement learning	6
7	Exams	3
	Total	42

Grading

Midterm Exams	20-40%
Homework	0-15%
Project	20-25%
Final	40%

Course Outcomes:

Students should be able to:

1. Understand the three main styles of learning: supervised, unsupervised, and reinforcement.
2. Apply simple algorithms for supervised, unsupervised and reinforcement learning.
3. Determine which of the three learning styles is appropriate to a particular problem domain.
4. Evaluate the performance of a simple learning system on a real-world dataset.

The relationship between Course Outcomes and Student Outcomes

Course Outcome	Student Outcomes					
	1	2	3	4	5	6
1.	H					
2.		H				
3.	M					
4.		M				H